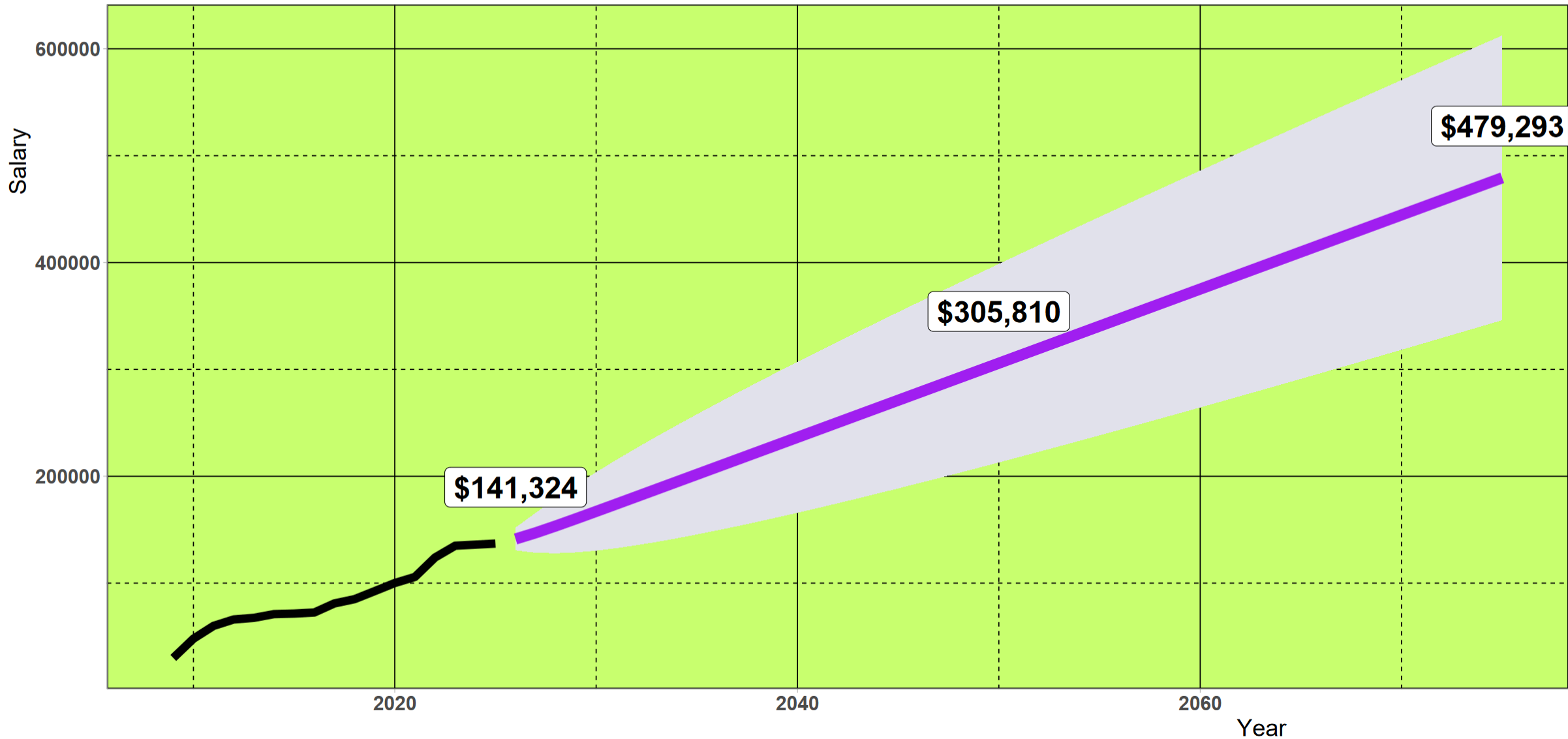


Salary Forecast Practice



```
# The following code to create a dataframe and remove duplicated rows is always executed and acts as a preamble for your script:
```

```
# dataset <- data.frame(Salary, Year)
# dataset <- unique(dataset)
```

```
# Paste or type your script code here:
```

```
options(scipen=999)
```

```
library(forecast)
```

```
library(tseries)
```

```
library(ggplot2)
```

```
salaryhistory <- dataset
```

```
#makes time series:
```

```
salaryhistorytimeline=ts(salaryhistory$Salary,start = min(salaryhistory$Year),end=max(salaryhistory$Year),frequency=1)
```

```
#this is where it picks the p, d, and q for the arima:
```

```
salaryfuturemodel=auto.arima(salaryhistorytimeline,ic="aic",trace=TRUE)
```

```
salaryhistorytimeline=ts(salaryhistory$Salary,start = min(salaryhistory$Year),end=max(salaryhistory$Year),frequency=1)
```

```
salaryfuturemodel=auto.arima(salaryhistorytimeline,ic="aic",trace=TRUE)
```

```
#95% confidence level and h = time horizon. 1 above = year so this is 50 years.
```

```
mysalaryforecast=forecast(salaryfuturemodel,level=c(95),h=50)
```

```
#need to get historic data
```

```
df_salaryhistory <- data.frame(
  Year = as.numeric(time(mysalaryforecast$x)),
  Salary = as.numeric(mysalaryforecast$x),
  Type = "Historical"
)
```

```
#do the forecast
```

```
df_forecastpart <- data.frame(
  Year = as.numeric(time(mysalaryforecast$mean)),
  Salary = as.numeric(mysalaryforecast$mean),
  Lower = as.numeric(mysalaryforecast$lower[, "95%"]),
  Upper = as.numeric(mysalaryforecast$upper[, "95%"]),
  Type = "Forecast"
)
```

```
#to make dataframe for datalabels. AND I copied this part from AI since I didn't know how to do it :-)
```

```
df_labels <- df_forecastpart[c(1, round(nrow(df_forecastpart)/2), nrow(df_forecastpart)), ]
```

```
#to make dataframe for datalabels. And I copied this part from AI since I didn't know how to do it .-)
```

```
df_labels <- df_forecastpart[c(1, round(nrow(df_forecastpart)/2), nrow(df_forecastpart)), ]
```

```
# 2. Add rounding to the Salary values so the labels look clean (e.g., $65,320 instead of $65320.123)
```

```
df_labels$LabelText <- paste0("$", format(round(df_labels$Salary, 0), big.mark = ","))
```

```
ggplot()+  
  geom_ribbon(data = df_forecastpart, aes(x = Year, ymin = Lower, ymax = Upper), fill = "#E3E3EE", linewidth = 0.3)+  
  geom_line(data = df_salaryhistory, aes(x = Year, y = Salary), size = 3, color = "black")+  
  geom_line(data = df_forecastpart, aes(x = Year, y = Salary), size = 4, color = "purple")+  
  ggtitle("Salary Forecast Practice")+  
  xlab("Year")+  
  ylab("Salary")+  
  theme_light()+  
  theme(  
    panel.background = element_rect(fill = "darkolivegreen1", color = "black", linewidth = 1),  
    plot.background = element_rect(fill = "white"),  
    plot.title = element_text(hjust = 0.5, size = 30),  
    axis.title.y = element_text(hjust = 0.8, size = 18),  
    axis.title.x = element_text(hjust = 0.8, size = 18),  
    axis.text.y = element_text(size=15, face = "bold"),  
    axis.text.x = element_text(size=15, face = "bold"),  
    panel.grid.major = element_line(color = "black", linewidth = 0.4),  
    panel.grid.minor = element_line(color = "black", linewidth = 0.4, linetype = "dashed")  
  )+  
  
  geom_label(  
    data = df_labels,  
    aes(x = Year, y = Salary, label = LabelText),  
    fontface = "bold",  
    color = "black",  
    vjust = -0.8,      # Nudges the label slightly above the line  
    size = 8           # Adjusts text size  
  )
```

```
... ..
```